Overview

Recap

- Classification of pdes:
  - static solution, boundary value (mostly elliptic)
  - time evolution, initial value (mostly parabolic, hyperbolic)
- Finite differences transform pdes into matrix equations
- Boundary conditions:
  Dirichlet: subtract boundary terms from source term
  Periodic: wrap finite differences around boundary
- Static solution problems may be solved directly
- Time evolution problems are solved iteratively
  - Use von Neumann analysis to study stability
  - FTCS: unstable if $\Delta t/a^2 > C$ (C depends on the problem)
  - BTCS: implicit scheme, always stable
  - Crank–Nicolson: implicit, second order in time, always stable

Relaxation methods for static boundary value problems

We want to sove $\mathcal{L}\Phi = \rho$, $\mathcal{L}$ = elliptic operator, eg. ∇^2

Start with initial guess, let system 'relax' $\longrightarrow$ solution of pde

Diffusion in computer time

$$\frac{\partial \Phi}{\partial t} = \mathcal{L}\Phi - \rho \qquad \mathcal{L} = \text{elliptic operator, eg. } \nabla^2$$

$\rightarrow$ stationary (late time) solution fulfils $\mathcal{L}\Phi = \rho$

We converted the problem to initial value (parabolic)

Common example: time-independent Schrödinger equation

$\hat{H}\Psi(x, y) = E\Psi(x, y)$ can be solved for smallest E by evolving

$$\frac{\partial}{\partial t}\Psi(x, y, t) = \hat{H}\Psi(x, y, t) \text{ to late times}$$

$\longrightarrow$ propagation in imaginary time

FTCS discretisation

$$\frac{\partial \Phi}{\partial t} = \frac{\Phi_{ij}^{(n+1)} - \Phi_{ij}^{(n)}}{\Delta t} = \nabla^2 \Phi - \rho$$

$$= \frac{\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n)} - 4\Phi_{ij}^{(n)}}{a^2} - \rho_{ij}$$

$$\Phi_{ij}^{(n+1)} = \left(1 - \frac{4\Delta t}{a^2}\right)\Phi_{ij}^{(n)} + \frac{\Delta t}{a^2}\left(\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n)}\right) - \Delta t \rho_{ij}$$

We saw last week that this is unstable if $\Delta t > a^2/4$

It slows down as Δt gets small

Relaxation method interpreted as Jacobi iteration

Relaxation as Jacobi iteration

Take biggest possible step: $\Delta t = a^2/4$

$$\Phi_{ij}^{(n+1)} = \frac{1}{4}\left(\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n)}\right) - \frac{a^2}{4}\rho_{ij}$$

Think of superscripts as iteration indices instead of as time discretization.

$\longrightarrow$ directly solving BVP ($\nabla^2\Phi = \rho$) by solving matrix equation.

Can speed up using Gauss-Seidel, SOR.

Summary

- Relaxation: convert static boundary value problem to initial value problem
  - start with some guess
  - solve diffusion equation in computer time
  - find stationary (late-time) solution
  - Can be interpreted as Jacobi iteration for solving by matrix inversion.